

Radioactive contamination

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	What is radioactive contamination?	The unwanted presence of radioactive atoms on or in other materials.	☐
2	Explain why radioactive contamination is dangerous.	The contaminating atoms keep emitting radiation for as long as they remain.	☐
3 ★	What is irradiation?	Exposing an object to nuclear radiation.	☐
4	Explain why irradiation is dangerous.	The radiation can ionise atoms in living cells, damaging or killing them.	☐
5	State whether irradiated objects become radioactive.	No: their nuclei do not become unstable.	☐
6	Give three precautions that protect people against irradiation.	Use shielding, keep your distance from the source, and limit the exposure time.	☐
7 ★	Explain why contamination by an alpha source inside the body is especially dangerous.	Alpha radiation is strongly ionising and is absorbed by nearby cells.	☐
8	State what the hazard from radioactive contamination depends on.	The type of radiation emitted.	☐
9 ★	Explain why studies of the effects of radiation on humans should be published.	So that other scientists can check the findings (peer review).	☐